

LaKeta L Kemp

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Education

- 2024-current University of California, San Diego (UCSD), PhD *Student: Data Science*, La Jolla, California
- 2023-2024 Howard University, *Master of Science: Data Science*, Washington, DC.
- 2023-2023 California Institute of Technology, Graduate Certificate, *Data Science*, Pasadena, California.
- 2022-2022 Massachusetts Institute of Technology, Graduate Certificate, *Applied Data Science*, Cambridge, Massachusetts.
- 1996-2008 Arizona State University, *Bachelor of Science: Molecular Biosciences*, Tempe, Arizona.

Patents and Selected Technical Publications

1. Bacterial identification. The present invention provides for separating bacterial species and serotypes using electrophoretic methods. T Taylor, M Hayes, LK Kemp, P Jones - US Patent 9,185,356, 2015
2. DC-iGDEP device for separating bacterial serotypes. T Taylor, M Hayes, LK Kemp, P Jones - US Patent 9,476,085, 2016
3. Paul V. Jones, Alexa F. DeMichele, LaKeta Kemp, and Mark A. Hayes. Differentiation of Escherichia coli Serotypes Using DC Gradient Insulator Dielectrophoresis. Anal. Bioanal. 2014, 406(1), 183192
4. Crowther CV, Hilton SH, Kemp L, Hayes MA. Isolation and identification of Listeria monocytogenes utilizing DC insulator-based dielectrophoresis. Analytica chimica acta. 2019 Aug 30;1068:41-51.

Employment Experience

- 2024-current Howard University, Center for Applied Data Science and Analytics Adjunct Data Scientist, Washington, DC
- 2024-2024 Lawrence Livermore National Laboratory, Graduate Intern: Critical Infrastructure Analysis, Cyber and Critical Infrastructure Summer Institute, Global Security Directorate, E Program, Livermore, California
- 2023-2024 Osmo Labs AI, Project Manager, Phoenix, Arizona
- 2023-2024 Advanced Degree Intern: Data Science, Salt River Project, Phoenix, Arizona

Selected Projects

- 2023 "TPE-XGBoost model with Tree-structured Parzen Estimator as a time-series forecasting model for Electric Load Forecasting."
- Detection of seasonal data using a discrete Fourier transform model with a Pearson Correlation Coefficient and MAPE metric to optimize the predictive capacity of electric load forecasting and decision-making for electric power reserves during high consumer demand to the utility-scale electric grid.
- 2023 "Short-term forecasting of daily precipitation using a SARIMA-XGBoost Model for forecasting rainfall data in Australia."
- Creation of a forecasting model that can predict the weather by combining historical patterns to predict the likelihood of future weather events. Severe weather is becoming increasingly important as human impact on the climate

increases. Confidently forecasting these extreme events is essential for protecting life.

- 2022 "Identification of Plasmodium spp. from digital blood smear slides using Computer Vision."
- Build and test a computer vision model that accurately analyzes digital images of blood smears, both infected and uninfected, for key features that identify malaria infections at the species level.

Current Certifications

- Microsoft Certified: Azure AI Fundamentals, Microsoft Issued Sep 2023, Credential ID E9FD47796C7E1EF5
- Human Subjects Research – Biomedical (Biomed) Foundations, CITI Program Issued Jul 2023 · Expires Jul 2024, Credential ID 56866011
- MIT Professional Education, Applied Data Science Program, Issued Jan 2023, Credential ID 66764790
- Ira A. Fulton Schools of Engineering at Arizona State University, Lean Six Sigma Yellow Belt (LSSYB), Issued Jun 2021, Credential ID 60d50a2b0348c77e12ee65cb

Professional Societies

UPE Upsilon Pi Epsilon Honor Society, Howard University Chapter, 2024
NEB Northeast Big Data Hub, 2024

Research Experience

2024-current Graduate Student Researcher, Climate Analytics Lab, UCSD, La Jolla, CA
I develop and train neural network-based emulators to serve as computationally efficient surrogates for complex Earth System Models (ESMs). This machine learning approach is designed to significantly accelerate climate projection simulations, reducing the time and resources required for climate analysis. The research contributes to improving the speed and accuracy of climate model outputs, thereby enhancing our understanding of the impacts of climate change.

2024-2024 Graduate Intern: Critical Infrastructure Analysis, Cyber and Critical Infrastructure Summer Institute, Lawrence Livermore National Laboratory, Global Security Directorate, E Program, Livermore, California
I conducted advanced geospatial and statistical analysis on high-dimensional data, contributing to the development of new simulation algorithms and optimization models. This work aimed to assess the reliability of critical power systems and enhance national emergency preparedness. The findings were then compiled into technical reports advising business partners on concrete strategies for hardening this vital infrastructure.

Major Awards and Honors

University of California, San Diego (UCSD) | *La Jolla, CA Doctor of Philosophy, Data Science*
| *September 2024 – Present*

- Honors: GEM Triton Scholar, Competitive Edge PhD Student Mentor

Howard University | *Washington, DC Master of Science, Data Science* | *May 2024*

- Honors: Mastercard Data Science Fellow, AIM-AHEAD Fellow, Upsilon Pi Epsilon Honor Society

Grants and Fellowships

2023-2024	Mastercard Data Science Impact Fellowship.
2024-2024	Artificial Intelligence/Machine Learning Consortium to Advance Health Equity and Researcher Diversity (AIM-AHEAD) All of Us Traineeship
2024-2025	The National GEM Consortium PhD Engineering and Science Fellowship

Teaching Experience

2024, 2025	Primary Instructor, Howard University Data Science for Social Impact Summer Bootcamp.
2024-current	Adjunct Data Scientist, Howard University, Center for Applied Data Science and Analytics, Washington, DC, DATA-200: Introduction to Data Science, DATA-801: Internship, DATA-802: Graduate Capstone.
2025-current	Instructional Assistant, University of California, San Diego, DSC 257R: Data Mining on the Web: Current Methods in Recommender Systems, Data Mining, and Predictive Analytics.

Workshops/Symposia Attended

Spring 2024	Women in Data Science Annual Conference, Stanford University, Palo Alto, California. March 5-6.
Spring 2024	Displacement and Reparation: Climate, Labor and Migration Justice Symposium, University of California, San Diego, La Jolla, California. February 7.
Summer 2025	Frontiers in Ocean Sciences Symposium, National Science Foundation, June 30.
Summer 2025	Communicating Science for Policy, National Science Policy Network, George Mason University, Fairfax, Virginia. July 21-30.

Contributions/Synergistic Activities

Coming soon. This is where the Médecins Sans Frontières project should go.

Research Interests

My primary research interest lies at the intersection of machine learning and climate science, with a focus on developing computationally efficient solutions to pressing environmental challenges. I am currently working on creating neural network-based emulators for complex Earth System Models (ESMs), a technique designed to drastically accelerate climate projection simulations. By building these fast and accurate surrogate models, my goal is to make high-resolution climate analysis more accessible. This forms the foundation for proactive tools, such as climate displacement early warning systems, that can better predict and mitigate the human impact of extreme weather events.

To ground these predictive models, a key part of my research involves the robust calibration of remote sensing data. I am particularly focused on developing methodologies to fuse and calibrate European Space Agency (ESA) satellite data with in-situ ground sensor readings, especially in regions disparately affected by climate extremes. Ensuring this data alignment is crucial for reducing uncertainty and enhancing the reliability of localized climate impact models, which is the necessary first step toward developing dependable real-world applications.

Crucially, my approach to developing these models is rooted in a commitment to ethical AI and social equity. I believe the communities most disproportionately affected by climate extremes mustn't be just subjects of the data but active participants in creating these solutions. This involves designing models that are transparent, fair, and directly address the needs and contexts of vulnerable populations. This ensures that the outputs do not perpetuate existing biases but instead empower inclusive and just climate adaptation strategies.

The ultimate application of this ethically-grounded research is to translate these predictive capabilities into actionable humanitarian intelligence. I am particularly motivated to further the mission of organizations like Médecins Sans Frontières (Doctors Without Borders) by developing weather anomaly systems that can provide critical foresight for their field operations. By forecasting impending floods, droughts, or heatwaves with greater lead time and geographic precision, these tools can enable the prepositioning of medical supplies, the proactive deployment of mobile clinics, and the anticipation of public health crises, directly supporting life-saving interventions in the world's most vulnerable communities.